Capturing Geopolitical Network Dynamics for Crude Oil Realized Volatility Forecasting: A Temporal Knowledge Graph Approach

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Abstract

Accurate volatility forecasting is essential for hedging and procurement planning in energy markets. However, standard geopolitical risk indices often compress heterogeneous global events into single scalar time series, discarding the topological structure and transmission pathways of international conflicts. In this paper, we introduce a spatio-temporal graph autoencoder that learns structured relational representations of the evolving global conflict network. This network is derived from a large-scale temporal knowledge graph recording interactions among producing countries, consuming nations, and maritime chokepoints. By design, our framework integrates these embeddings as conditioning signals for multi-horizon realized volatility forecasting through a task-decoupled learning objective. This strategy insulates the geopolitical structure learning from downstream financial supervision, reducing the impact of scale asymmetry between high-frequency global event streams and sparse financial targets.

Diebold–Mariano tests indicate a clear horizon-dependent pattern. At the daily horizon, implied volatility (OVX) remains competitive, while the proposed framework delivers statistically significant QLIKE reductions over the HAR-OVX-GPR baseline from one week onward. These gains are visible at all longer horizons and are most pronounced during episodes of elevated market stress, including the COVID-19 shock, the Russia-Ukraine War, and the recent US-Iran war. Ablation studies further indicate that the network-based representation contributes incremental predictive information relative to scalar aggregation, particularly at intermediate and longer horizons.

Keywords

Realized volatility, Geopolitical risk, Graph neural networks, Temporal knowledge graphs, Representation learning, Rolling window estimation

1 Introduction

Crude oil realized volatility is a key indicator for global energy markets, with direct implications for derivative pricing, risk management, and macroeconomic policy. The Heterogeneous Autoregressive (HAR) model [8] has become the standard benchmark for realized volatility forecasting, and a growing body of literature has incorporated geopolitical risk proxies—most prominently the GPR index of [6]—as exogenous variables within this framework. Yet existing proxies reduce the inherently relational nature of international tensions to a scalar time series, discarding structural information that may be relevant for longer-horizon forecasting, including which actors are involved, their positions within the global energy supply network, and the pathways through which tensions propagate. For market participants such as refinery risk managers, sovereign energy funds, and central banks, these longer-horizon forecasts are often the ones most closely tied to planning and hedging decisions.

Capturing this structure requires representing the conflict network as a Temporal Knowledge Graph (TKG), in which nodes correspond to geopolitical actors and directed, typed edges represent time-stamped dyadic events [19, 26]. Unlike scalar indices that treat all events as interchangeable counts, this representation preserves who is involved, what type of interaction occurs, and how tensions evolve over time—dimensions that become increasingly relevant at longer forecasting horizons, where scalar indices have shown limited incremental value [2, 10]. Graph representation learning methods—in particular R-GCN [25] and its temporal extensions—can encode these evolving structures into dense embeddings that preserve both topological and temporal information. However, this approach faces a fundamental scale asymmetry: the GDELT Project records a vast stream of dyadic interactions over many years, whereas realized volatility provides only a limited sequence of daily observations. To address this mismatch, we adopt an online learning procedure that updates the geopolitical encoder sequentially as new daily information arrives, while keeping the financial forecasting task decoupled from the structure-learning stage. This design allows the representation model to be trained under a realistic information arrival process and aligns the methodology with operational out-of-sample forecasting constraints.

Evaluation methodology warrants careful attention in this setting. The financial forecasting literature has repeatedly documented that in-sample predictive power frequently fails to translate into out-of-sample gains [28], and that the choice of loss function critically determines model rankings in volatility settings [24]. We therefore adopt a strict rolling retrain protocol updated daily, consistent with the standards advocated for operational volatility forecasting [2]. The chosen benchmark—HAR-OVX-GPR—is deliberately demanding: OVX encodes the forward-looking expectations of options markets, making it difficult to surpass at short horizons; combined with GPR, this specification provides an informationally rich baseline against which incremental predictive content can be assessed.

We propose a spatio-temporal graph autoencoder that formalizes the daily conflict network as a TKG over energy-critical nodes spanning major producing countries, key consuming nations, and strategically important maritime chokepoints. The encoder combines a spatial R-GCN layer that aggregates typed relational information across the conflict graph with a temporal GRU layer that propagates node-level memory across successive daily snapshots, together constituting the core spatio-temporal engine. Auxiliary components—a lightweight global refinement pass and a daily activity-count injection—provide complementary signals without altering the fundamental representational mechanism. Trained via a self-supervised link reconstruction objective under an online protocol, the resulting embeddings are reduced via PCA and integrated into a HAR-X model, remaining operationally tractable by offloading event extraction to GDELT's upstream infrastructure.

Our main contributions are as follows:

- We introduce a *structural geopolitical risk factor* derived from temporal knowledge graph representation learning as an exogenous variable for multi-horizon crude oil realized volatility forecasting, offering a principled alternative to frequency-based scalar indices. Unlike scalar decomposition approaches that attempt to recover structure post-hoc from an already-compressed signal, our framework preserves relational topology from the ground up.
- We propose a spatio-temporal graph autoencoder comprising a spatial R-GCN layer, a temporal GRU layer, a global Transformer refinement pass, and an activity-count late fusion module, trained via a self-supervised link reconstruction objective that strictly decouples geopolitical structure learning from financial supervision and addresses the scale asymmetry between decade-long event streams and limited volatility observations.
- We implement a rigorous daily-retrain rolling protocol and document a horizon-dependent pattern: structural embeddings provide no incremental gains at the daily horizon, where HAR-OVX-GPR retains primacy, but yield statistically significant improvements from one week onward, with the gains becoming more pronounced at longer horizons. This indicates that topological geopolitical information is especially valuable when forecasting windows are long enough for conflict structure to matter.
- We build an end-to-end, real-world *strict out-of-sample* (OOS) forecasting pipeline that mirrors operational constraints: a daily batch job executed after the New York market close ingests only information available up to the close, performs one-step online updates of the geopolitical encoder, refits rolling HAR-X models, and emits multi-horizon forecasts under a no-look-ahead protocol suitable for reproducible backtesting.

2 Related Work

A substantial body of empirical research has established that financial asset return volatility exhibits pronounced clustering and long-memory properties, rendering it meaningfully predictable from its own history [5, 13]. The Heterogeneous Autoregressive (HAR) model operationalizes this observation by decomposing realized volatility into daily, weekly, and monthly components, parsimoniously capturing the multi-scale persistence that characterizes high-frequency financial data [8]. By exploiting high-frequency realized measures rather than daily returns alone, HAR-type models have been shown to deliver more accurate out-of-sample forecasts than standard GARCH specifications across a range of assets and evaluation settings [1, 8].

The forecasting literature has since explored numerous extensions to the baseline HAR framework. Lasso-based regularization effectively captures a linear footprint of volatility dynamics [3]. Machine learning algorithms have subsequently been shown to capture nonlinear dynamics in crude oil futures volatility more effectively than linear models, reducing forecast errors in regimes characterized by structural breaks and tail events [22]. Sentiment and attention indicators derived from financial news improve equity volatility forecasts, with gains varying systematically across attention regimes and market conditions [4]. Graph neural networks capturing volatility spillover effects across assets—by explicitly modeling the topological structure of inter-market relationships—yield meaningfully superior forecasting performance relative to models that treat each asset independently [29].

Options-derived forward-looking information has proven particularly valuable for energy markets. The CBOE Crude Oil Volatility Index (OVX) embeds the market's own assessment of near-term uncertainty, and incorporating it into HAR specifications achieves significant improvements in short-horizon accuracy [18], confirmed across multiple methodological settings [16, 23]. The HAR-OVX specification has consequently emerged as the de facto benchmark, combining historical persistence with market-implied expectations in a parsimonious and empirically validated framework [9, 10]. The choice of evaluation metric also matters critically: given the non-Gaussian, heavy-tailed distribution of realized volatility, the QLIKE loss function provides more reliable model rankings than MSE-based criteria, particularly under the extreme conditions most relevant to risk management [24].

Beyond scalar risk indices, disaggregating aggregate uncertainty metrics along their constituent dimensions has been shown to enhance oil-market predictability [17]. Decomposing the GPR index into geopolitical threats and acts yields superior performance at intermediate and long horizons, particularly with nonlinear estimators [17], and replacing aggregate U.S. economic-policy uncertainty with state-level counterparts produces further improvements in machine-learning-based realized variance forecasts [7]. However, all such approaches share a fundamental limitation: they decompose top-down from an already-aggregated scalar index, and information destroyed during the original compression cannot be recovered by subsequent decomposition. Our approach differs in direction: rather than disaggregating a compressed index after the fact, we construct a high-resolution geopolitical representation from the bottom up, assembling hyper-granular dyadic events directly into a relational structure and bypassing the scalar bottleneck entirely.

This bottom-up assembly is structured simultaneously along three axes—*who* is involved (energy-critical nodes), *what type* of interaction occurs (CAMEO-coded relation types), and *how* tensions evolve over time (temporal graph dynamics)—preserving the complex relational topology that traditional indices implicitly and irreversibly discard.

These findings collectively paint a landscape in which machine learning methods, textual sentiment, and network-based spillover models each offer informative signals [3, 4, 29]. Yet this richness conceals a fundamental evaluation problem. Chassot and Audrino [2] provide large-scale empirical evidence bearing directly on this issue, conducting a systematic comparison of volatility forecasting models across a dataset of 1,445 individual stocks. Their central finding is that forecasting performance is highly sensitive to the fitting scheme employed: specifically, a conventional linear HAR model configured with sufficiently large estimation windows and subjected to a daily re-estimation rolling-window protocol consistently proves difficult to surpass. Under the QLIKE loss function, many complex nonlinear machine learning models fail to produce statistically significant incremental predictive gains over this parsimonious linear benchmark when the evaluation protocol is held constant and operationally realistic [2]. This result underscores that the apparent advantage of sophisticated models documented in prior literature may, at least in part, reflect differences in fitting schemes rather than genuine improvements in predictive content.

Consequently, studies seeking to surpass HAR-type models under these strict operational constraints typically require additional inputs capturing jump dynamics or large sets of ML-selected predictors, often achieving only marginal and inconsistent improvements [12, 20, 22]. This operational difficulty is magnified when incorporating deep structural architectures. Standard dynamic graph learning paradigms, such as DyRep [26] and RE-NET [19], are fundamentally designed for offline batch optimization, requiring multiple training epochs and computationally expensive Backpropagation Through Time (BPTT) over entire historical sequences. In a financial context demanding continuous model re-estimation, executing a full parameter retraining from scratch on a daily basis becomes operationally intractable due to the severe computational burden of processing decade-long streaming event data daily.

Against this backdrop, the present study adopts the strictest available evaluation standard: QLIKE loss [24], Diebold–Mariano significance testing [11], rolling estimation with daily retraining [2], and explicit comparison against OVX- and GPR-augmented baselines [10, 17] across seven forecasting horizons. The adoption of daily rolling retraining is directly motivated by the evidence in [2]: by ensuring that the HAR-OVX-GPR baseline is always estimated with a sufficiently large and continuously updated window, the evaluation design places the linear benchmark in its empirically strongest configuration, yielding an informationally rich and competitive point of comparison against which any incremental contribution of topological geopolitical features can be assessed without confounding attributable to suboptimal baseline calibration.

To reconcile the computational tension between structural graph representation learning and the operational necessity of daily rolling re-estimation, this study is the first to reformulate temporal knowledge graph representation into a decoupled online learning paradigm tailored for financial forecasting. By insulating the structural feature extraction from downstream financial supervision and executing strict one-step online gradient updates upon the arrival of daily streaming snapshots, our framework accommodates the rigorous evaluation standards advocated by [2] without suffering from the computational bottlenecks inherent to full offline batch reconstruction from scratch.

While recent work has adopted similarly rigorous standards [2, 10], the integration of topological geopolitical features within such a framework remains entirely unexplored. Existing studies typically satisfy only subsets of these criteria—rigorous rolling-window evaluation without topological representations [2, 10], or topological modeling without the strict operational protocol considered here [29]. This gap motivates both our evaluation design and our claim that the reported improvements represent genuine, operationally meaningful forecasting gains.

3 Preliminaries

3.1 Realised Volatility and the HAR-X Model

Daily realised volatility is the sum of squared intraday returns over M valid 1-minute bars:

$$\mathrm{RV}_t = \sum_{\tau=1}^{M} r_\tau^2. \tag{1}$$

Multi-day backward averages $\overline{\mathrm{RV}}_{t-k+1:t} = k^{-1}\sum_{j=0}^{k-1}\mathrm{RV}_{t-j}$ for $k \in \{1, 5, 22\}$ define the HAR model [8] (where $\overline{\mathrm{RV}}_{t:t} \equiv \mathrm{RV}_t$):

$$\overline{\mathrm{RV}}_{t+1:t+h} = \alpha + \beta_d \mathrm{RV}_t + \beta_w \overline{\mathrm{RV}}_{t-4:t} + \beta_m \overline{\mathrm{RV}}_{t-21:t} + \varepsilon_{t+h}, \tag{2}$$

where $\overline{\mathrm{RV}}_{t+1:t+h} = h^{-1}\sum_{k=1}^{h}\mathrm{RV}_{t+k}$ is the mean realised volatility over the next h trading days. The HAR-X extension augments (2) with exogenous regressors $\mathbf{x}_t \in \mathbb{R}^q$:

$$\overline{\mathrm{RV}}_{t+1:t+h} = \alpha + \beta_d \mathrm{RV}_t + \beta_w \overline{\mathrm{RV}}_{t-4:t} + \beta_m \overline{\mathrm{RV}}_{t-21:t} + \boldsymbol{\gamma}^\top \mathbf{x}_t + \varepsilon_{t+h}, \tag{3}$$

where $\mathbf{x}_t$ comprises $\log \mathrm{OVX}_t$ and the geopolitical scalar z_t (Section 4.4). All specifications are estimated via Ridge regression to mitigate overfitting at longer horizons:

$$\hat{\boldsymbol{\theta}}^{\mathrm{Ridge}} = \arg\min_{\boldsymbol{\theta}} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2 \right\}, \tag{4}$$

with λ held fixed uniformly across all models and horizons to ensure consistent shrinkage throughout the rolling estimation windows.

3.2 Temporal Knowledge Graph

Definition 3.1 (Temporal Knowledge Graph). A *temporal knowledge graph* (TKG) is a sequence $\mathcal{G}_{1:T} = (\mathcal{G}_1, \ldots, \mathcal{G}_T)$, where each snapshot $\mathcal{G}_t = (\mathcal{E}, \mathcal{R}, \mathcal{F}_t)$ shares fixed entity set $\mathcal{E}$ and relation-type set $\mathcal{R}$, and $\mathcal{F}_t \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$ is the set of typed directed edges on day t.

In our instantiation, $|\mathcal{E}| = 45$ energy-critical nodes and $|\mathcal{R}| = 11$ conflictual CAMEO top-level categories (`CAMEO` ≥ 10) [15]; each snapshot is built from GDELT [21] events whose subject and object both belong to $\mathcal{E}$.

3.3 Relational Graph Convolutional Network

An R-GCN [25] aggregates neighbour features per relation type, with a distinct weight matrix $\mathbf{W}_r^{(l)}$ for each $r \in \mathcal{R}$:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\mathbf{W}_0^{(l)} \mathbf{h}_i^{(l)} + \sum_{r\in\mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} \mathbf{W}_r^{(l)} \mathbf{h}_j^{(l)} \right), \tag{5}$$

where $\mathcal{N}_i^r$ is the set of neighbours of node i under relation r and $c_{i,r} = |\mathcal{N}_i^r|$ is a normalisation constant.

3.4 Self-Supervised Graph Autoencoder

A graph autoencoder pairs an encoder $\mathbf{M}^{(t)} = f_\theta(\mathcal{G}_t) \in \mathbb{R}^{|\mathcal{E}|\times d_T}$ with an MLP decoder scoring continuous edge weights:

$$\hat{w}_{sro}^{(t)} = f_\phi\Big([\, \mathbf{m}_s^{(t)} \| \mathbf{e}_r \| \mathbf{m}_o^{(t)} \,] \Big), \quad f_\phi : \mathbb{R}^{3d_T} \to \mathbb{R}. \tag{6}$$

Training minimises a weighted reconstruction loss over balanced positive and negative triples:

$$\mathcal{L}_{\mathrm{recon}} = \frac{1}{|\mathcal{T}^+| + |\mathcal{T}^-|} \sum_{(s,r,o,w)\,\in\,\mathcal{T}^+\cup\mathcal{T}^-} \left(\hat{w}_{sro}^{(t)} - w \right)^2, \tag{7}$$

Table 1: Taxonomy of the $|\mathcal{E}| = 45$ energy-critical nodes. Country ISO-3 codes denote sovereign states; virtual nodes (prefix underscore) capture non-state and multilateral actors.

Category	Role	N	Nodes (ISO-3 / Virtual)
Major Producers	OPEC/Non-OPEC & swing producers	17	SAU, IRQ, IRN, ARE, KWT, QAT, NGA, LBY, VEN, GUY, COL, RUS, USA, CAN, NOR, MEX, BRA
Chokepoint Sovereigns	Maritime transit control	9	EGY, DJI, SGP, IDN, PAN, OMN, TUR, VNM, PHL
Conflict Hotspots	Risk-channel transmission	8	UKR, ISR, PSE, SYR, LBN, TWN, PAK, YEM
Major Consumers	Demand-side pressure	7	CHN, IND, DEU, FRA, NLD, GBR, AUS
Non-State Actors	Multilateral & systemic risk	4	_OPEC, _IEA, _TERROR, _INTL_ORG
Total		**45**	

where $w = \log(1+n_{sro})$ for observed edges and $w = 0$ for corrupted negatives. Crucially, $\mathcal{L}_{\text{recon}}$ uses no financial supervision, decoupling geopolitical representation learning from the downstream volatility regression and resolving the scale asymmetry identified in Section 4.

4 Methodology

This section presents the proposed framework for extracting geopolitical structure signals from event data and integrating them into a volatility forecasting model. Figure 1 provides an overview of the complete pipeline. The framework consists of four components: (i) construction of a temporally evolving geopolitical knowledge graph (§4.1); (ii) a spatio-temporal encoder comprising relational graph convolution and gated recurrence to capture topological and temporal dynamics (§4.2); (iii) a self-supervised online training protocol that updates model parameters strictly one day at a time to prevent look-ahead bias (§4.3); and (iv) a downstream HAR-X forecasting regression that identifies the most severe geopolitical vulnerability via max-pooling and incorporates it alongside established volatility regressors (§4.4).

4.1 Geopolitical Graph Construction

The GDELT Project monitors global news media in real-time, extracting dyadic interactions and translating them into a hyper-granular stream of machine-readable events [21]. Rather than applying a destructive compression to aggregate these events into a single daily volume or sentiment scalar, we assemble them bottom-up into a daily directed multigraph $\mathcal{G}_t = (\mathcal{E}, \mathcal{R}, \mathcal{F}_t)$, preserving the topological transmission pathways of geopolitical shocks that scalar indices inherently discard.

Energy-critical node selection. To filter the computationally intractable global event stream into a financially relevant topology, we define a fixed entity set $\mathcal{E}$ of $|\mathcal{E}| = 45$ energy-critical nodes. As detailed in Table 1, these nodes are organised into five functional categories to cover the complete supply-chain topology of the global crude oil market.

By explicitly separating producing nations from transit chokepoints and active conflict zones, this taxonomy ensures that a geopolitical shock near the Strait of Hormuz is topologically distinct from a diplomatic protest in Western Europe, a distinction that any scalar aggregation irrecoverably destroys. The selection rationale is grounded in three empirically supported mechanisms:

- **Supply and demand core.** The 17 producing nations collectively hold over 80% of global proven crude oil reserves; Russia and the United States are included as swing producers whose geopolitical posture exerts significant causal influence on prices. China and India, the world's first and third largest importers, anchor the demand side; their bilateral energy relationships with Russia and the Gulf states constitute a structurally distinct geopolitical dimension [14].
- **Transit vulnerability.** The nine chokepoint sovereigns control the world's primary maritime oil corridors: the Strait of Hormuz (OMN, IRN) carried 20.9 mb/d in 2023, representing over one-quarter of global seaborne oil trade, while the Strait of Malacca (SGP, IDN) recorded 23.7 mb/d, the highest of any single chokepoint [27].
- **Risk-channel transmission.** Active conflict nodes and non-state actors are explicitly modelled because their shock dynamics transmit to oil prices through the risk premium channel [14]. The ECB documents that geopolitical events originating in Israel raise Brent crude prices by 0.8–1.5% within five trading days [14].

Relation type mapping. Each directed edge $(s, r, o) \in \mathcal{F}_t$ is drawn from the Conflict and Mediation Event Observations (CAMEO) coding scheme [15], which provides a four-level hierarchical taxonomy of interstate interactions ranging from cooperative to conflictual. We retain only the eleven conflictual top-level categories (`CAMEO` ≥ 10), spanning from Demand (10) through Use Unconventional Mass Violence (20), yielding $|\mathcal{R}| = 11$ relation types. Cooperative interactions (`CAMEO` < 10) are excluded on the grounds that crude oil volatility is primarily driven by conflict-induced supply disruption and risk-premium shocks rather than diplomatic accommodation [6]. By aggregating at the top level, we capture the distinct topological semantics of geopolitical friction while maintaining a tractable relational space.

Snapshot construction. Daily snapshot $\mathcal{G}_t$ is constructed by aggregating all GDELT events whose subject and object both belong to $\mathcal{E}$, aligned to the NYMEX WTI crude oil futures trading calendar. Specifically, each event timestamp (`DATEADDED`, in UTC) is first converted to `America/New_York` time and assigned to a *base calendar date* using a shifted day boundary at 17:45 ET; this base date is then mapped to the next valid NYMEX energy trading day using the CME Globex Energy calendar, ensuring that all edges in $\mathcal{G}_t$ are strictly observable before the corresponding trading session begins. Edge weights are defined as $\log(1 + n_{srot})$, where n_{srot} denotes the event count of type r between actors s and o within the window. The 45-node subgraph yields between 150,000 and 300,000 raw events per day over the full sample (April 2015–March 2026), which are aggregated into weighted daily (s, r, o) interactions. On non-trading days, no snapshot is constructed; on trading days with no recorded events for a given entity pair—comprising fewer than 2% of the sample—the corresponding edge weight is set to zero, consistent with the assumption that the structural state of the geopolitical network does not change discontinuously in the absence of recorded events.

4.2 Spatio-Temporal Graph Encoder

Given daily snapshot $\mathcal{G}_t$ and memory matrix $\mathbf{M}_{t-1} \in \mathbb{R}^{N\times d_T}$, the encoder produces updated memory $\mathbf{M}_t$ through a three-stage pipeline of spatial aggregation, temporal recurrence, and global refinement (Figure 1; $d_S = 16$, $d_T = 32$).

Relational Graph Convolution. We adopt an R-GCN [25] to aggregate neighbour features separately for each of the $|\mathcal{R}| = 11$ conflictual relation types, with a distinct weight matrix $\mathbf{W}_r$ per relation. Node i's spatial feature is computed as:

$$\tilde{\mathbf{h}}_i^{(t)} = \mathrm{ReLU}\left(\mathbf{m}_{i,t-1}\mathbf{W}_{\mathrm{self}} + \sum_{r\in\mathcal{R}}\sum_{j\in\mathcal{N}_r(i,t)} \mathbf{m}_{j,t-1}\mathbf{W}_r\right),$$

yielding $\tilde{\mathbf{H}}_t \in \mathbb{R}^{N\times d_S}$. A learnable projection $\mathbf{W}_{\mathrm{proj}} \in \mathbb{R}^{d_S\times d_T}$ with dropout ($p = 0.1$) maps this to d_T, denoted as $\hat{\mathbf{h}}_i^{(t)}$.

Gated Recurrent Update. The projected feature and inherited memory are fused node-wise by a GRU cell to produce the day-t intermediate node embedding:

$$\mathbf{m}_{i,t}^{\mathrm{raw}} = \mathrm{GRU}\left(\hat{\mathbf{h}}_i^{(t)},\ \mathbf{m}_{i,t-1}\right),$$

whose gates learn to balance accumulated geopolitical memory against immediate localized structural changes, yielding the matrix $\mathbf{M}_t^{\mathrm{raw}} \in \mathbb{R}^{N\times d_T}$.

Global Transformer and Late Fusion. To refine these localized, node-level temporal representations into a coherent systemic network state, the output of the GRU cell is passed to a global Transformer layer. The multi-head self-attention mechanism computes long-range dependencies across all N energy-critical nodes, allowing the framework to capture how localized friction implicitly shifts the broader geopolitical equilibrium. The cross-node interaction is formalised as:

$$\mathbf{M}_t^{\mathrm{trans}} = \mathrm{LayerNorm}\left(\mathbf{M}_t^{\mathrm{raw}} + \mathrm{MultiHeadAttention}(\mathbf{M}_t^{\mathrm{raw}})\right).$$

To incorporate ambient event intensity without altering the structural representations, this attention-refined matrix is combined with the daily activity-count matrix $\mathbf{c}_t$ via a late fusion module, yielding the finalized daily memory matrix:

$$\mathbf{M}_t = \mathrm{Tanh}\left(\mathrm{Fusion}(\mathbf{M}_t^{\mathrm{trans}},\ \mathbf{c}_t)\right) \in \mathbb{R}^{N\times d_T}.$$

The resulting matrix $\mathbf{M}_t$ is detached before propagation to day $t + 1$.

Memory initialisation. $\mathbf{M}_0 \sim \mathcal{N}(0,\ 0.01^2\mathbf{I})$. The model undergoes four sequential warm-up passes over April 2015–December 2017 (chronological order strictly preserved) before the evaluation window opens on 1 January 2018.

4.3 Self-Supervised Online Training

The encoder is trained without oil-market supervision via a *weighted link prediction* pretext task. For each day t, positive triples are $\mathcal{T}_t^+ = \{(s, r, o, \log(1 + n_{srot}))\}$ and a decoder MLP scores each triple:

$$\hat{w}_{srot} = f_\theta\left(\left[\mathbf{m}_s^{(t)} \| \mathbf{e}_r \| \mathbf{m}_o^{(t)}\right]\right), \quad f_\theta : \mathbb{R}^{3d_T} \to \mathbb{R}. \tag{8}$$

The MSE loss over balanced positive and negative triples ($|\mathcal{T}_t^-| = |\mathcal{T}_t^+|$) is:

$$\mathcal{L}_t = \frac{1}{|\mathcal{T}_t^+| + |\mathcal{T}_t^-|}\sum_{\substack{(s,r,o,w)\,\in\\ \mathcal{T}_t^+\cup\mathcal{T}_t^-}} (\hat{w}_{srot} - w)^2, \tag{9}$$

with $w = 0$ for negatives; MSE on log-counts differentiates event intensities more naturally than binary cross-entropy. Negatives are formed by corrupting subject or object with a closed-world collision check.

From 1 January 2018, the model operates in a strict online regime: read day-t triples, take one gradient step on $\mathcal{L}_t$, update $\mathbf{M}_t$, emit embeddings—no future information is ever accessed. Gradient norms are clipped to 0.5; Adam with lr $= 5\times10^{-4}$, weight decay 10^{-4}.

GDELT 2.0 updates at 15-minute intervals, so each day-t snapshot is finalised 15 minutes before the NYMEX open, ensuring $\mathbf{M}_t$ is fully observable before the corresponding trading session and no look-ahead bias is introduced.

4.4 Downstream Volatility Forecasting

HAR baselines. Let RV_t be the daily realised variance of WTI crude oil futures computed from 1-minute returns, and $\overline{\mathrm{RV}}_{t-k:t}$ its k-day backward average. The HAR model [8] is defined as:

$$\begin{aligned}\overline{\mathrm{RV}}_{t+1:t+h} &= \alpha + \beta_d\mathrm{RV}_t + \beta_w\overline{\mathrm{RV}}_{t-4:t}\\ &\quad + \beta_m\overline{\mathrm{RV}}_{t-21:t} + \varepsilon_{t+h}.\end{aligned}$$

The primary benchmark HAR-OVX-GPR further adds $\log\mathrm{OVX}_t$ and GPR_t [6]:

$$\begin{aligned}\overline{\mathrm{RV}}_{t+1:t+h} &= \alpha + \beta_d\mathrm{RV}_t + \beta_w\overline{\mathrm{RV}}_{t-4:t}\\ &\quad + \beta_m\overline{\mathrm{RV}}_{t-21:t} + \beta_{\mathrm{OVX}}\log\mathrm{OVX}_t\\ &\quad + \beta_{\mathrm{GPR}}\mathrm{GPR}_t + \varepsilon_{t+h}.\end{aligned} \tag{10}$$

HAR-X: proposed specification. The encoder emits the node embedding matrix $\mathbf{M}_t \in \mathbb{R}^{N\times d_T}$. To aggregate the cross-sectional risk features across the entire network topology, we compute the element-wise arithmetic mean across all N energy-critical nodes:

$$\mathbf{z}_t = \frac{1}{N}\sum_{i=1}^{N}\mathbf{m}_{i,t} \in \mathbb{R}^{d_T},$$

where the arithmetic mean is calculated element-wise across the entity dimension. This formulation compresses the localized node-level features into a systemic representation, allowing the network-wide geopolitical friction to be captured continuously. By utilizing an arithmetic mean, the downstream specification preserves the aggregate structural uncertainty profile refined by the upstream global Transformer layer, ensuring that the cross-sectional dynamics of the entire energy network are symmetrically represented before matrix compression.

A strict rolling PCA (window = 1,000 days, refitted daily on past data only) compresses $\mathbf{z}_t$ to a scalar $z_t \in \mathbb{R}$ representing the primary axis of systemic geopolitical risk. The proposed model replaces GPR with z_t:

$$\begin{aligned}\overline{\mathrm{RV}}_{t+1:t+h} = \alpha + \beta_d \mathrm{RV}_t + \beta_w \overline{\mathrm{RV}}_{t-4:t} \\ + \beta_m \overline{\mathrm{RV}}_{t-21:t} + \beta_{\mathrm{OVX}} \log \mathrm{OVX}_t \\ + \beta_z z_t + \varepsilon_{t+h}.\end{aligned} \tag{11}$$

All coefficients are estimated via Ridge regression on a rolling window of 1,000 observations, re-estimated daily, with a parsimonious regularisation penalty applied uniformly across all models and horizons to ensure consistent shrinkage and prevent in-sample overfitting at longer forecast windows.

Horizons and evaluation. Forecasts span five horizons: 1D, 1W (h=5), 2W (h=10), 3W (h=15), and 1M (h=22). All models are evaluated by QLIKE:

$$\mathrm{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{\mathrm{RV}_{t+h}}{\hat{\sigma}^2_{t+h}} - \log\frac{\mathrm{RV}_{t+h}}{\hat{\sigma}^2_{t+h}} - 1\right),$$

a consistent loss under the proportionality assumption [24]. Statistical significance is assessed with the Diebold–Mariano test [11] using Newey–West HAC standard errors.

5 Experiments

5.1 Experimental Setup

Data. The empirical implementation utilizes high-frequency financial metrics combined with granular planetary event streams. Daily realized variance (RV_t) for the West Texas Intermediate (WTI) crude oil futures (the front-month continuous contract traded on the NYMEX) is systematically constructed from 1-minute clean intraday returns filtered over the full trading session [1]. The historical horizon spans from April 2015 through April 2026, encompassing approximately 2,700 trading sessions [10, 21]. The macroeconomic benchmarks, namely the CBOE Crude Oil Volatility Index (OVX) and the Caldara and Iacoviello Geopolitical Risk (GPR) index [6], are synchronized at a daily frequency, with missing values attributable to weekend or holiday mismatches handled via standard forward-filling protocols.

The strict out-of-sample (OOS) evaluation window officially opens on 1 January 2018 and extends through 28 April 2026, yielding an expansive testing sample of T = 2,094 energy trading days [2, 10]. This evaluation sequence is chronologically preceded by a warm-up period running from April 2015 through December 2017, which is reserved exclusively for model initialization and parameter estimation. By extending the OOS evaluation boundary to late April 2026, the dataset also covers the period of heightened geopolitical stress associated with the escalation in the Middle East, including maritime disruptions in critical energy chokepoints and military confrontation involving the United States, Israel, and Iran from late 2024 into early 2026 [17, 27]. This period provides a demanding setting for assessing the capacity of the proposed framework to accommodate abrupt shifts in geopolitical and market conditions.

On the unstructured data processing side, the raw global event pipeline is derived from the GDELT 2.0 database, containing over 570 million international records. Following domain-specific actor filtering and virtual-node taxonomy mapping, a refined set of approximately 90 million energy-critical events is retained. These are subsequently aggregated bottom-up into over 17 million daily directional bilateral–event-code snapshots [15]. Because the standard GPR index is released with a reporting lag and at a lower temporal frequency, we deliberately configure the competitive HAR-OVX-GPR baseline as an *oracle*-style benchmark by providing it with contemporaneous daily values that are not available to live market practitioners [10]. Consequently, any statistically significant predictive outperformance achieved by the proposed framework over this benchmark establishes a conservative lower bound on the economic and operational gains that may be available in real-world deployments.

Baselines and evaluation. To isolate the incremental predictive contribution of the learned topological geopolitical representations, we benchmark the proposed **HAR-X** framework (formulated via the HAR-OVX + RollPCA(1) specification in Eq. 11) against a hierarchical suite of established empirical baselines: the pure historical persistence **HAR** model [8], the options-implied expectation model **HAR-OVX** [18], and the macro-index augmented oracle baseline **HAR-OVX-GPR** [10]. All parameter spaces are estimated via Ridge regression over a strict rolling window fixed at 1,000 daily observations, with model parameters re-estimated daily to mirror operational out-of-sample forecasting constraints [2]. For reproducibility, an anonymised repository provides code, configuration files, and additional baselines evaluated under the same rolling protocol[1].

Consistent with practical risk management boundaries, the forecasting horizons are restricted to five discrete steps: 1D, 1W (h=5), 2W (h=10), 3W (h=15), and 1M (h=22) [2, 17]. The primary statistical evaluation criterion is governed by the QLIKE loss function [24], which has been shown to provide robust model rankings even when realized volatility is measured with noise. Pairwise predictive significance is formally determined via the Diebold–Mariano (DM) testing protocol [11]. Given that multi-step forecasting horizons ($h > 1$) induce an $\mathrm{MA}(h-1)$ error structure through overlapping prediction windows, the associated variance estimation is adjusted using Newey–West heteroskedasticity and autocorrelation consistent (HAC) standard errors with a parsimonious lag choice [11].

A fundamental hazard in the use of deep temporal architectures in financial backtesting pipelines is data leakage and look-ahead bias. To preserve forecasting integrity, an end-to-end point-in-time (PiT) chronological alignment pipeline is implemented in the ingestion process, establishing a non-overlapping cutoff before the

[1]Code and data are available at: https://anonymous.4open.science/r/Crude-Oil-Realized-Volatility-Forecasting-A-Temporal-Knowledge-Graph-Approach-50E0/

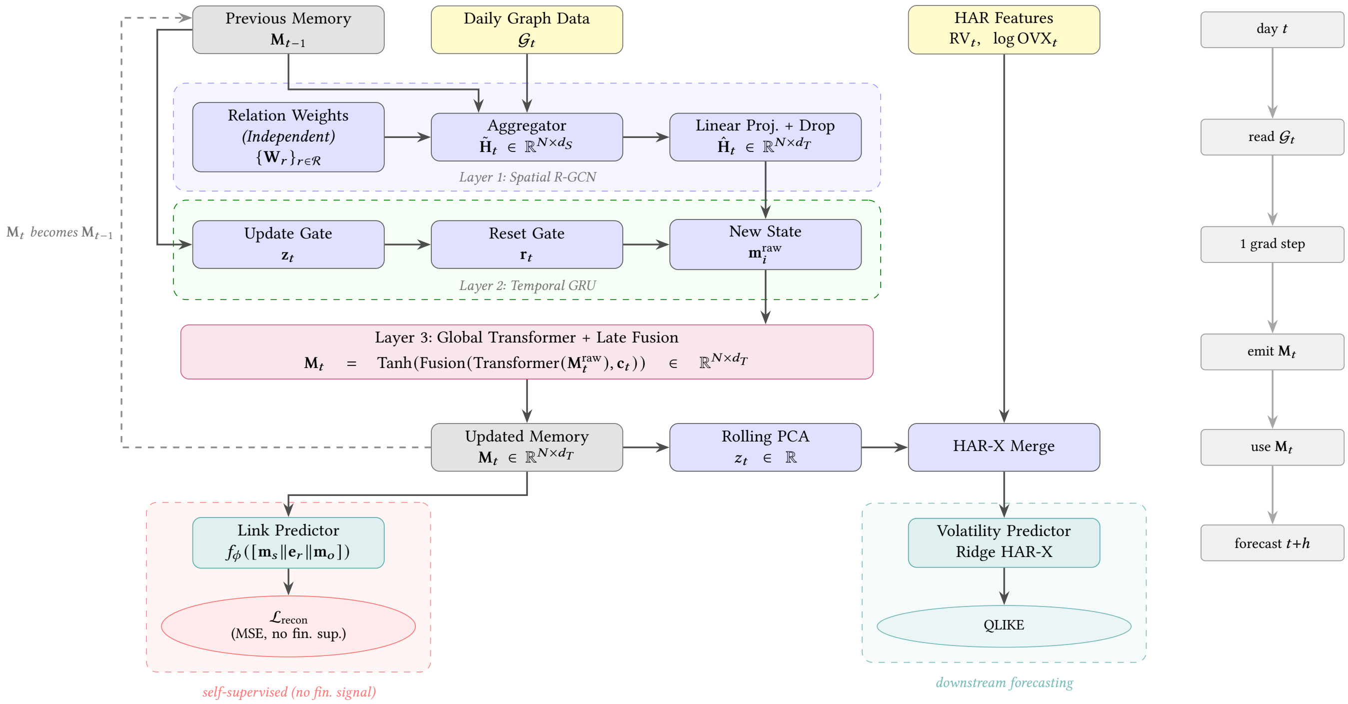

Figure 1: Proposed spatio-temporal graph autoencoder. The encoder stacks three hierarchical layers (Spatial R-GCN → Temporal GRU → Global Transformer) followed by a late fusion mechanism. The self-supervised link-prediction head ($\mathcal{L}_{\mathrm{recon}}$) receives no financial supervision, decoupling geopolitical representation learning from the downstream Ridge HAR-X forecasting head (QLIKE). Right panel: strict online protocol.

Table 2: Point-in-Time (PiT) Alignment Pipeline to avoid look-ahead bias.

NY Time (ET)	CME WTI Market State	Data Pipeline Action	Index
Day T 17:00	Session Closes	Compute Day T RV; record closing market expectations (OVX_t) and oracle indices (GPR_t).	Day T
Day T 17:44	Market Closed	Ingest the final streaming GDELT event snapshot for the daily geopolitical multigraph.	Day T
Day T 17:45	Market Closed (Pre-open)	**Spatio-temporal graph cutoff**; execute one-step online parameter update and emit memory embedding $\mathbf{M}_t$.	Day T
15-Minute Operational Forecasting Window: Formulate Exogenous Scalar z_t and Execute Ridge HAR-X Model			
Day T 18:00	Day T+1 Session Opens	Begin accumulating physical Day T+1 high-frequency realized volatility (Target boundary).	Day T+1

official opening of each corresponding energy trading session (Table 2).

5.2 Forecast Accuracy

Table 3 reports the out-of-sample QLIKE losses across the five forecast horizons considered in the main analysis. At the **1D horizon**, HAR-OVX-GPR attains the lowest nominal loss (0.1241), while the proposed model remains close at 0.1263. The DM test indicates no statistically significant difference at this horizon, suggesting that the two specifications are observationally similar in the very short run. At **1W**, the proposed model achieves the lowest QLIKE (0.1404), marginally improving on HAR-OVX-GPR (0.1436). The advantage becomes more pronounced at **2W** and **3W**, where the proposed model delivers the lowest losses in both cases (0.1711 and 0.1956, respectively), indicating that the learned topological features begin to contribute more visibly at intermediate horizons. At **1M**, the proposed specification continues to outperform the baselines, recording a QLIKE loss of 0.2366 relative to 0.2406 for HAR-OVX-GPR and 0.3184 for HAR. Overall, the results suggest that the incremental predictive content of the proposed framework is more discernible beyond the shortest horizon, while remaining competitive even at the daily frequency.

Table 3: Out-of-sample QLIKE loss across forecast horizons. Lower is better. Bold = minimum loss at each horizon. Superscripts denote pairwise DM significance of the proposed model relative to HAR-OVX-GPR: $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$, NS = not significant.

Horizon	HAR	HAR-OVX	HAR-OVX-GPR	Proposed
1D	0.1577	0.1274	**0.1241**	0.1263
1W	0.1703	0.1428	0.1436	**0.1404*****
2W	0.2154	0.1725	0.1731	**0.1704*****
3W	0.2571	0.1976	0.1978	**0.1951*****
1M	0.3184	0.2407	0.2406	**0.2366*****

5.3 Temporal Stability: Regime-Dependent Performance

Figure 2 presents the time-varying loss differential between the proposed model and the HAR-OVX-GPR benchmark over the full out-of-sample period across five forecast horizons. Each panel corresponds to one horizon and plots realized volatility as a black line, while the shaded area indicates the sign of the smoothed QLIKE loss differential: green regions correspond to intervals in which the proposed model delivers lower loss than HAR-OVX-GPR, whereas red regions indicate the opposite. This representation emphasizes the temporal clustering of relative forecasting performance rather than aggregating it into a single summary statistic. To provide context for these regime shifts, the onset dates of major geopolitical and market shocks—specifically the COVID-19 pandemic declaration, the Russia-Ukraine War, and the US-Iran war—are marked in each panel.

Across horizons, the proposed model exhibits a regime-dependent pattern that is broadly consistent with the temporal variation identified in the statistical tests. At the shortest horizon (1D), relative performance is mixed, with intervals in which HAR-OVX-GPR remains competitive. A notable feature of the 1D panel is that, although red regions dominate for much of the sample, the proposed model does outperform HAR-OVX-GPR during the more recent US-Iran war episode. At longer horizons, the proposed model appears more frequently associated with lower losses during periods of elevated volatility. This is visible in the broader green regions surrounding the COVID-19 episode, the outbreak of the Russia-Ukraine War in 2022, and the geopolitical disturbances observed during the US-Iran war in 2026. These patterns are consistent with the interpretation that the proposed topological features become more informative when geopolitical and market conditions undergo abrupt adjustments.

The figure also indicates that the relative performance differential varies over time rather than remaining constant. The proposed specification tends to perform better during high-volatility regimes, while the gap narrows during calmer periods. In the intermediate and monthly horizons, the shaded areas remain predominantly green across extended intervals, including much of 2021 and 2023–2025. This suggests that the RollPCA compression preserves useful predictive information over time, while maintaining performance that remains stable across both turbulent and relatively tranquil market conditions.

5.4 Ablation Study

To isolate the sources of predictive power, we evaluate four ablated variants: (i) **Simple Count**—replaces the full GNN with a raw conflictual-event volume scalar computed over the same 45-node subgraph, testing whether targeted topological filtering alone drives gains without relational aggregation; (ii) **w/o GRU**—processes each daily snapshot independently without sequential memory, isolating the contribution of temporal recurrence; (iii) **w/o Transformer**—retains the temporal update mechanism but removes self-attention-based contextual mixing, testing whether cross-event aggregation within the update block adds incremental value; and (iv) **Full Model**—the proposed RGCN + GRU + Max Pooling architecture.

Table 4: Ablation study: out-of-sample QLIKE loss. Bold = minimum at each horizon.

Horizon	HAR-OVX-GPR	Simple Count	w/o GRU	w/o Transformer	Full Model
1D	0.1241	**0.1231**	0.1249	0.1248	0.1263
1W	0.1436	**0.1421**	0.1432	0.1430	0.1404
2W	0.1731	0.1715	0.1729	0.1727	**0.1704**
3W	0.1978	0.1961	0.1976	0.1974	**0.1951**
1M	0.2406	0.2390	0.2415	0.2413	**0.2366**

6 Conclusion

We proposed a spatio-temporal graph autoencoder that recovers the relational topology of geopolitical risk discarded by scalar indices, and integrates the resulting representation as a structured exogenous input into a HAR-X model for crude oil realized volatility forecasting. Evaluated under a strict daily-retrain rolling protocol, the framework exhibits a clear horizon-dependent pattern: OVX retains primacy at the daily horizon, while the structural embedding delivers statistically significant improvements at horizons beyond one week, with gains becoming more pronounced at intermediate and longer forecast horizons. In particular, the model performs better during episodes of elevated geopolitical stress, including the COVID-19 disruption, the Russia-Ukraine War, and the US-Iran war, suggesting that relational information becomes more informative when conflict dynamics intensify.

Ablation evidence indicates that the predictive contribution is associated with the learned network structure rather than with raw event volume alone, and that the self-supervised online learning procedure is useful for aligning geopolitical representation learning with the sequential arrival of financial data. By decoupling structure learning from downstream forecasting, the framework also addresses the scale asymmetry between hyper-granular event streams and comparatively scarce volatility observations. Future work may extend the framework to multi-commodity spillover networks or incorporate LLM-derived sentiment into edge weights to further refine geopolitical risk transmission.

GenAI Usage Disclosure

During the preparation of this work, the authors used ChatGPT (GPT-4o) to assist with the refinement of English prose, the correction of grammatical errors, and the translation of specific technical

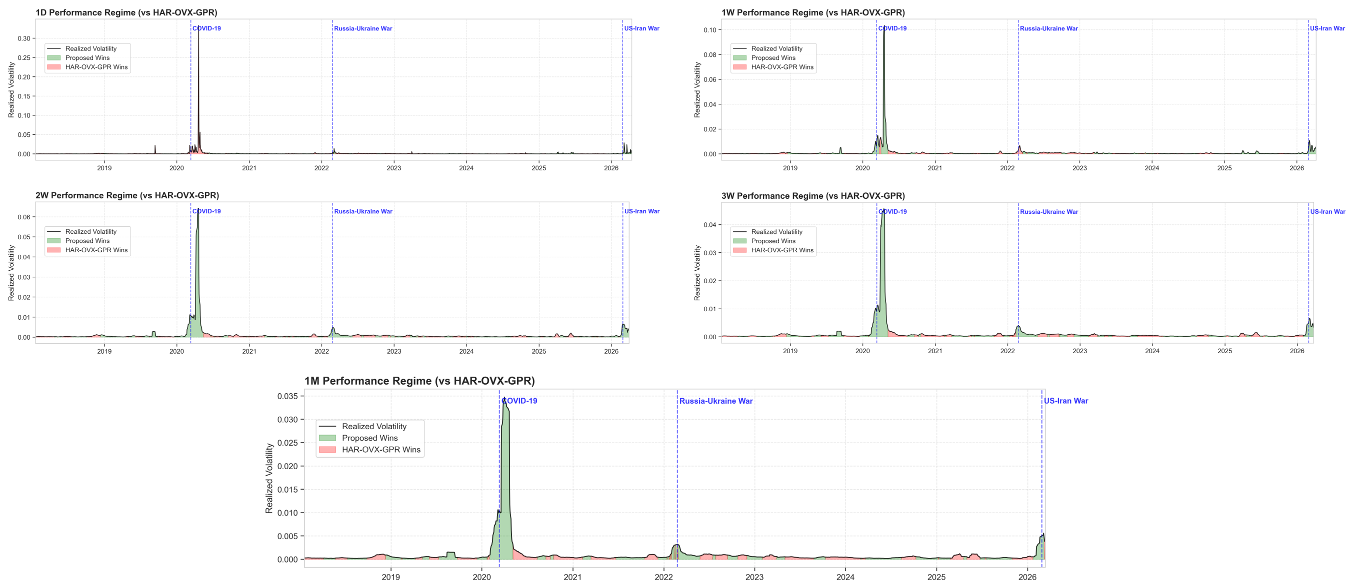

Figure 2: Regime-dependent performance of the proposed model relative to HAR-OVX-GPR across five forecast horizons. Each panel plots realized volatility (black line) together with the sign of the smoothed QLIKE loss differential. Green shading indicates intervals in which the proposed model achieves lower loss than HAR-OVX-GPR, while red shading indicates the opposite. Vertical markers denote the onset dates of the COVID-19 pandemic declaration, the Russia-Ukraine War, and the US-Iran war.

descriptions. Furthermore, this tool was utilized to aid in the drafting and debugging of Python code snippets used for data processing and model evaluation. After the tool-assisted generation, the authors reviewed and edited the content as needed and take full responsibility for the content and accuracy of the final published work.

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